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Week 2 Assignment - Probability

Review Learning Objectives

Problem Set Up

Let X denote the temperature at which a certain chemical reaction takes place. Suppose the pdf (probability density function) of X , denoted by $f(x)$, has the following form: $f(x) = \begin{cases} \frac{1}{9}(4 - x^2) & \text{if } x \in [-1, 2] \\ 0 & \text{otherwise} \end{cases}$. What is the probability that you've learned before starting the assignment? I'm here to help.

$f(x) = \begin{cases} \frac{1}{9}(4 - x^2) & \text{if } x \in [-1, 2] \\ 0 & \text{otherwise} \end{cases}$ [Let's chat](#)

It might help you to get started with this problem to sketch $f(x)$.

Due Oct 2, 11:59 PM IST 3 left (5 attempts every 24 hours)

Attempts 0/5

Try again 10 / 10 points

1. Determine the cdf (cumulative distribution function), denoted by $F(x)$, if $x < -1$.

0

Correct

Your grade To pass you need at least 70%. We keep your highest score: 90%

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2. Determine the cdf (cumulative distribution function), denoted by $F(x)$, if $x \in [-1, 2]$. Your answer will be a mathematic expression.

0 / 10 points

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$$-\frac{5}{9} + \frac{2}{9} + \frac{11}{9}$$
$$1/27*(127*x*x^3+11)$$

Incorrect Think about the cdf formula and try again:

$$F[X] = \Pr[X \leq x] = \int_{-\infty}^x f(x)dx$$

3. Determine the cdf (cumulative distribution function), denoted by $F(x)$, if $x > 2$. Your answer will be an integer.

10 / 10 points

1

Correct Think about the cdf formula and try again:

$$F[X] = \Pr[X \leq x] = \int_{-\infty}^x f(x)dx$$

4. Find the mean of X .

10 / 10 points

0.25

Correct

$$E[X] = \int_{-1}^2 xf(x)dx = \int_{-1}^2 \frac{1}{9}(4 - x^2)dx = \frac{1}{4}$$

5. Find the variance of X .

10 / 10 points

- ☐ 1/4
- ☐ 3/5
- ☐ 9/25
- ☒ 43/80

Correct Your answer is correct! The correct process is as below,

$$E[X^2] = \int_{-1}^2 xf(x)dx = \int_{-1}^2 \frac{1}{9}(4 - x^2)dx = \frac{1}{4}$$
$$E[X^2] = \int_{-1}^2 x^2f(x)dx = \int_{-1}^2 \frac{1}{9}(4 - x^2)dx = \frac{3}{5}$$
$$\text{var}[X] = E[X^2] - E[X]^2 = \frac{43}{80}$$

6. Let $X_1, X_2, \dots, X_n$ be independent and identically distributed (i.i.d.) copies of X .

30 / 30 points

Find the cdf of the following random variable: $L_n = \min\{X_1, X_2, \dots, X_n\}$.
(Hint: Let $G(x) = \Pr(L_n \leq x)$. Using the definition of L_n and the fact that X_i 's are independent, write $G(x)$ as the product of simple terms that you have determined in the previous parts.)

- ☐ $G(x) = [1 - F(x)]^n$
- ☒ $G(x) = 1 - [1 - F(x)]^n$
- ☐ $G(x) = F(x)^n$
- ☐ $G(x) = 1 - F(x)$

Correct

$$G[X] = \Pr(L_n \leq x) = \Pr(\min(X_1, X_2, \dots, X_n) \leq x) = 1 - \Pr(\min(X_1, X_2, \dots, X_n) > x) = 1 - \prod_{i=1}^n \Pr(X_i > x) \text{ [Independence]} = 1 - \Pr(X_1 > x)^n \text{ [Identical]} = 1 - [1 - \Pr(X_1 \leq x)]^n = 1 - [1 - F(x)]^n$$

7. Let $X_1, X_2, \dots, X_n$ be independent and identically distributed (i.i.d.) copies of X .

20 / 20 points

Find the cdf of the following random variable: $U_n = \max\{X_1, X_2, \dots, X_n\}$.

- ☒ $F(x)^n$
- ☐ $F(x)$
- ☐ $F(x)^2$

Correct

$$U[X] = \Pr(U_n \leq x) = \Pr(\max(X_1, X_2, \dots, X_n) \leq x) = \prod_{i=1}^n \Pr(X_i \leq x) \text{ [Independence]} = \Pr(X_1 \leq x)^n \text{ [Identical]} = F(x)^n$$